

UID LAB EXPERIMENT - 2

1.Command Line Interface(CLI):

CLI (Command Line Interface)

CLI implementation where users can add, view, and remove tasks using the terminal.

```
tasks = []
def add_task(task):
    tasks.append(task)
    print(f'Task '{task}' added.")

def view_tasks():
    if tasks:
        print("Your tasks:")
        for idx, task in enumerate(tasks, 1):
            print(f'{idx}. {task}')
    else:
        print("No tasks to show.")

def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        print(f'Task '{removed_task}' removed.")
    else:
        print("Invalid task number.")

def main():
    while True:
        print("\nOptions: 1.Add Task 2.View Tasks 3.Remove
```

Task 4.Exit")

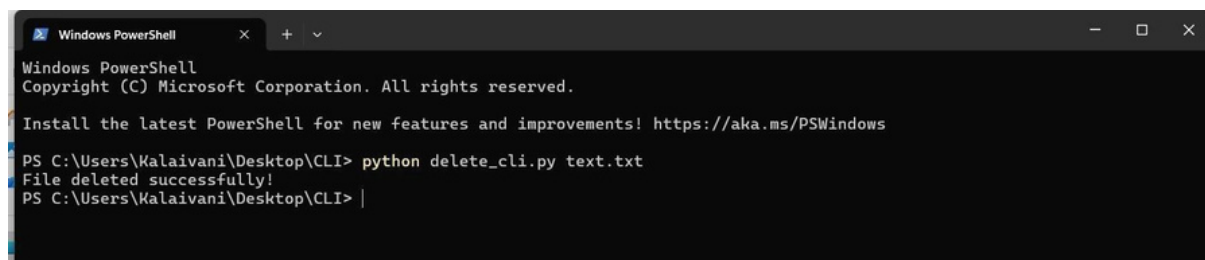
```
choice = input("Enter your choice: ")
if choice == '1.': task = input("Enter
task: ") add_task(task) elif choice ==
'2.': view_tasks() elif choice == '3':
task_number = int(input("Enter task number to
remove: "))
```

```
remove_task(task_number)
elif choice == '4': print("Exiting...")
break else: print("Invalid choice. Please
tryagain.")
```

```
if __name__ == "__main__":
```

```
main()
```

OUTPUT:



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Kalaivani\Desktop\CLI> python delete_cli.py text.txt
File deleted successfully!
PS C:\Users\Kalaivani\Desktop\CLI> |
```

2.Graphical User Interface:

Program:

Tkinter to create a simple GUI for our To-Do List application.

```
import tkinter as tk from tkinter import messagebox tasks = []
```

```
def add_task():
```

```
task = task_entry.get() if
task:
tasks.append(task)
task_entry.delete(0, tk.END)
update_task_list() else:
messagebox.showwarning("Warning", "Task cannot be
```

```
empty") def
update_task_list():
task_list.delete(0, tk.END) for
task in tasks:
task_list.insert(tk.END, task)
```

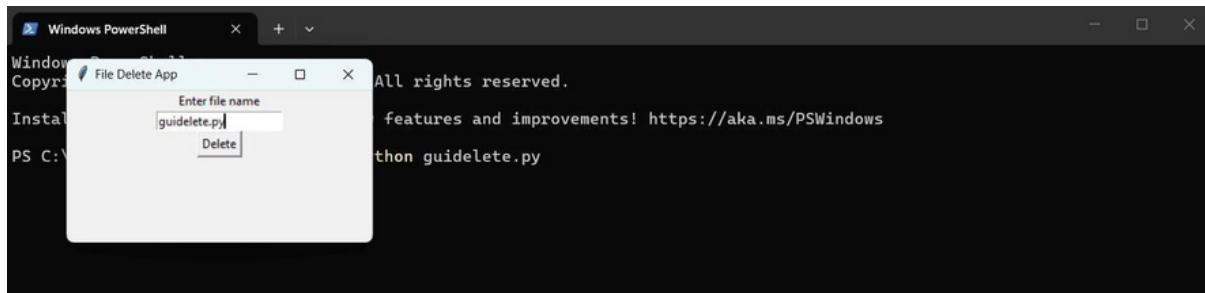
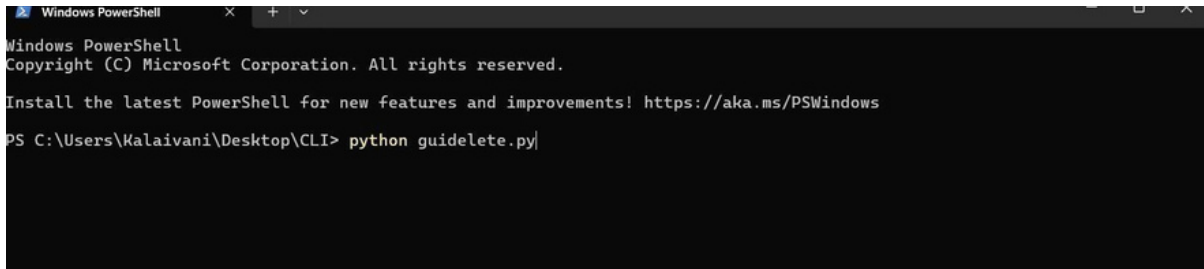
```
def remove_task():
selected_task_index = task_list.curselection() if
selected_task_index:
task_list.delete(selected_task_index)
tasks.pop(selected_task_index[0])
```

```
app = tk.Tk() app.title("To-Do List")
task_entry = tk.Entry(app, width=40)
task_entry.pack(pady=10)
```

```
add_button = tk.Button(app, text="Add Task",
command=add_task) add_button.pack(pady=5)
remove_button = tk.Button(app, text="Remove Task",
command=remove_task)
remove_button.pack(pady=5) task_list =
```

```
tk.Listbox(app, width=40, height=10)
```

```
task_list.pack(pady=10) app.mainloop() OUTPUT:
```



3. Voice User Interface:

Program: speech_recognition library for voice input and the pytsx3 library for text-to-speech output. Make sure you have these libraries installed (pip install SpeechRecognition pytsx3). import speech_recognition as sr import pytsx3 tasks = [] recognizer = sr.Recognizer()

```
engine = pytsx3.init() def
```

```
add_task(task):
```

```
tasks.append(task)
```

```
engine.say(f"Task {task} added")
```

```
engine.runAndWait() def
```

```
view_tasks(): if tasks:
```

```
engine.say("Your tasks are")
```

```
for task in
```

```
engine.say(task)
```

```
else:
```

```

engine.say("No tasks to show")
engine.runAndWait()
def
remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        engine.say(f"Task {removed_task} removed")
    else:
        engine.say("Invalid task number")
engine.runAndWait()
def recognize_speech():
    with sr.Microphone() as source:
        print("Listening...")
        audio = recognizer.listen(source)
    try:
        command = recognizer.recognize_google(audio)
    except sr.UnknownValueError:
        engine.say("Sorry, I did not understand that")
    engine.runAndWait()
    return None

```

```
def main():
```

```
    while True:
```

```
        engine.say("Options: add task, view tasks, remove
```

```
task, or exit")
```

```
        engine.runAndWait()
        command
```

```
        = recognize_speech()
        if not
```

```
        command: continue
        if "add
```

```
task" in command:
```

```
            engine.say("What is the task?")
```

```
            engine.runAndWait()
            task =
```

```
            recognize_speech()
```

```
            if task: add_task(task)
```

elif "view tasks" in command:

view_tasks()

elif "remove task" in command:

engine.say("Which task number to remove?")

engine.runAndWait() task_number =

recognize_speech() if task_number:

remove_task(int(task_number))

elif "exit" in command:

engine.say("Exiting...")

engine.runAndWait()

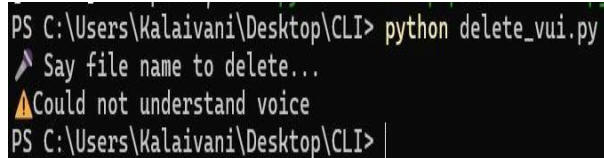
break else:

engine.say("Invalid option. Please try again.")

engine.runAndWait() if name == "main":

main()

OUTPUT:



```
PS C:\Users\Kalaivani\Desktop\CLI> python delete_vui.py
Say file name to delete...
⚠ Could not understand voice
PS C:\Users\Kalaivani\Desktop\CLI> |
```